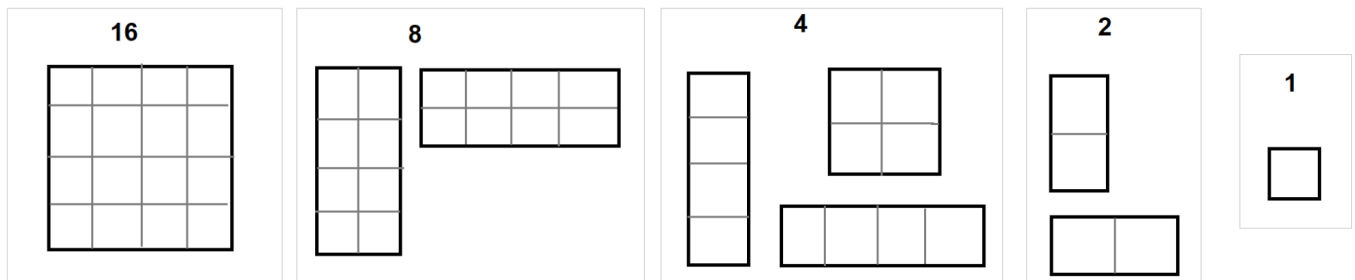


Lecture 5: Karnaugh Map or Veitch Diagram

Steps:

1. Identify the number of variables and draw the map.
2. Put 1 in designated cells
3. Make groups of adjacent 1's. For this, we have the following rules:
 - a. Try to make the largest group. [16,8,4,2,1]
 - b. Group size should be 2^n [16,8,4,2,1]. Group shape should be



- c. All 1 must be in groups.
 - d. You can cover a 1 multiple times. Only if it is needed for a bigger group to cover any uncovered 1's.
 - e. In case of don't care (X), we can use them to create larger groups. But we will not create groups of X only.
4. Check if there are any unnecessary groups by removing one group at a time and verifying whether all 1s are still covered by the remaining groups. If all 1s are still covered, the removed group is unnecessary.
5. From the remaining groups, derive the minimized function by finding common literals.
- 6.

Find the simplified equation: (4 variables)

		CD			
		C'D'	C'D	CD	CD'
AB	A'B'				
	A'B				
	AB				
	AB'				

		CD			
		C'D'	C'D	CD	CD'
AB	A'B'				
	A'B				
	AB				
	AB'				

AB \ CD				
	C'D'	C'D	CD	CD'
A'B'				
A'B				
AB				
AB'				

AB \ CD				
	C'D'	C'D	CD	CD'
A'B'				
A'B				
AB				
AB'				

AB \ CD				
	C'D'	C'D	CD	CD'
A'B'				
A'B				
AB				
AB'				

AB \ CD				
	C'D'	C'D	CD	CD'
A'B'				
A'B				
AB				
AB'				

AB \ CD				
	C'D'	C'D	CD	CD'
A'B'				
A'B				
AB				
AB'				

AB \ CD				
	C'D'	C'D	CD	CD'
A'B'				
A'B				
AB				
AB'				